APPENDIX TO "DECIDING WITHOUT GENERATING: WHEN DOES A STANDALONE DECISION MODEL PAY OFF IN LLM SYSTEMS?"

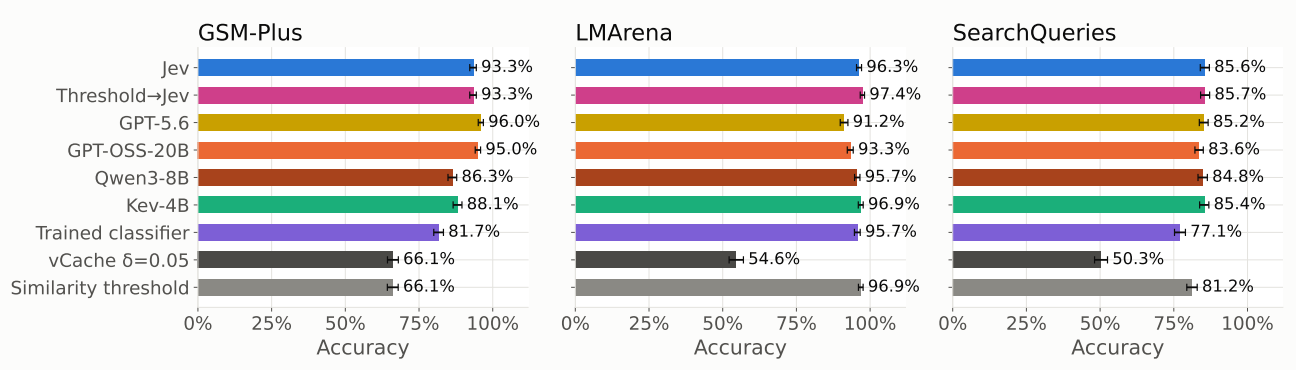


Figure A1. Cache-guard accuracy per decider and dataset with 95% CIs clustered by class. vCache as in Table 4 of the main paper.

A LABEL CORRECTION FOR THE TWO VCACHE DATASETS

On our first run, every LLM decider reached only 70–72% on SearchQueries, and Jev and the frontier model 71% on LMArena, against 93–97% for the hosted deciders on GSM-Plus. Samples of the disagreements put the fault in the labels: "best workout plans for women" and "best workout plan for women" were labeled different, while "count blank cells" and "count non blank cells" were labeled the same.

In SearchQueries, classes are transitive closures of pairwise LLM judgments within embedding clusters. That is valid only if "same" is a strict equivalence, and three things break it. (1) A cached answer is written for its own query: a general query may be judged close enough to two specific ones, but the answer stored for one specific query is wrong for the other. (2) Tolerance accumulates along a chain of individually acceptable links. (3) One mistaken link merges two whole classes, labelling pairs no model ever compared.

In LMArena, a class is "generated from the same seed prompt". An LLM wrote the variants and nothing checked that a variant still asks the same thing; some reverse a premise or change an exam code or a number. In the other direction, identical prompts that came from different seeds are labeled different (37 of the 2,600 pairs).

Correction. We label the pair directly: the new query and the cached query it would be served from. A written guideline, fixed in advance, defines the same request strictly (rewording is the same; any change of entity, number, direction, task, scope or output constraint is different). A frontier LLM (Claude Opus 5.5) judged all 5,200 dev and test pairs blind, seeing neither the original label nor the similarity nor any decider's output. Pairs where a confident verdict disagreed with the original label were judged again, blind, by a separate instance, and a label changed only when both agreed with high confidence. This changed 156 of 2,600 LMArena labels (6.0%) and 409 of 2,600 SearchQueries labels (15.7%). On the corrected labels, the first run's SearchQueries accuracy rose by 10 to 14 points for every LLM decider and for the similarity threshold.

Limits. The corrector is an LLM, and no person checked its labels, so the corrected labels may sit closer to an LLM decider's idea of the same request than human labels would. It judged every pair, not only those where deciders disagreed with the label, so that which pairs were judged did not depend on any decider. GSM-Plus labels are exact numeric answers and were not touched.

B ADDITIONAL RESULTS

Figure A1 gives the cache-guard accuracies of Table 4 with their confidence intervals, Figures A2–A4 add detail to the decision-only replay of Section 4, and Figure A5 to the in-loop arms.

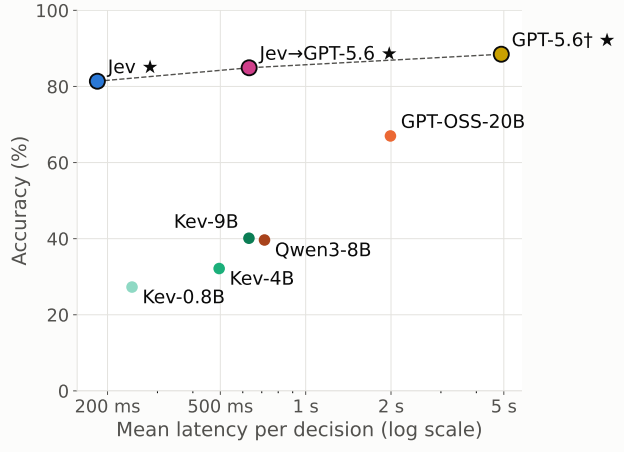


Figure A2. Decision-only replay: accuracy against mean latency per decision (log scale). Stars mark the accuracy–latency frontier; † as in Table 2.

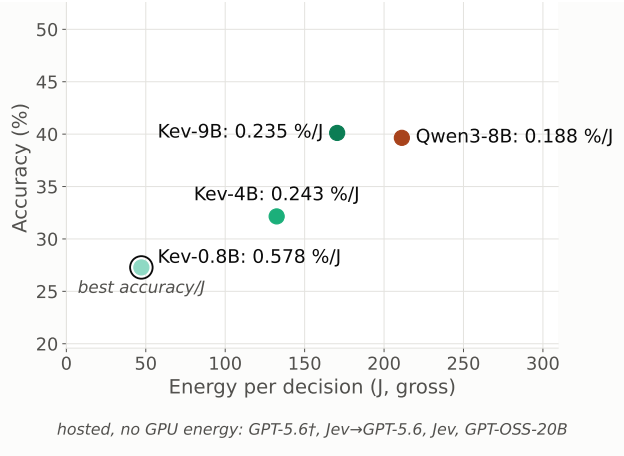


Figure A3. Decision-only replay: accuracy against gross GPU energy per decision for the local deciders on one A100-80GB. Point labels give accuracy points per joule (%/J).

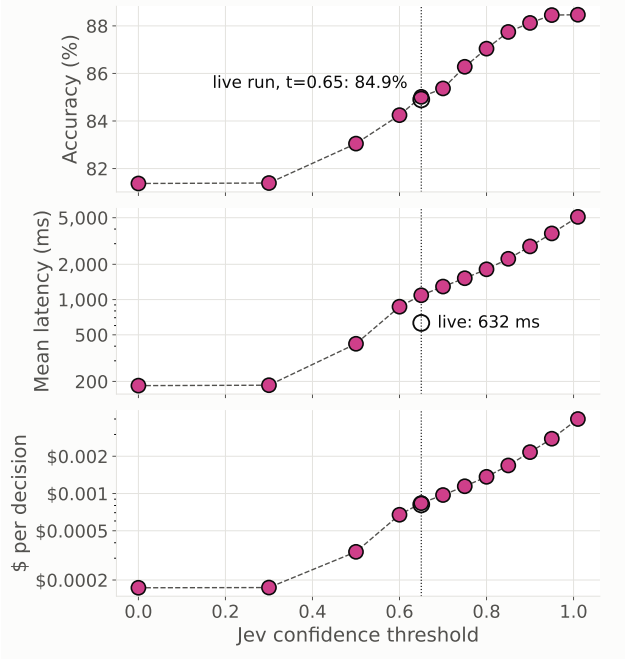


Figure A4. The Jev → GPT-5.6 cascade as its confidence threshold varies (escalate below it). Points are simulated from the standalone replays (latency and cost on log scales), the open ring is the live run at the chosen threshold of 0.65, and the dotted line marks that threshold.

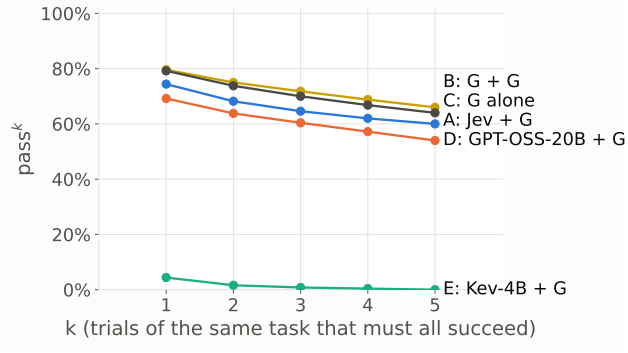


Figure A5. In-loop reliability: pass^k, the chance that k trials of the same task all succeed, per arm.